Smart Agricultural Technology 14 (2026) 102145

Contents lists available at ScienceDirect

Smart Agricultural Technology

journal homepage: www.journals.elsevier.com/smart-agricultural-technology

Developing and integrating trust modeling into multi-objective reinforcement learning for intelligent agricultural management

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ARTICLE INFO

Keywords:
Agricultural management
Reinforcement learning
Trust model
Human-AI interaction

ABSTRACT

Precision agriculture, enhanced by artificial intelligence (AI), offers promising tools like remote sensing, intelligent irrigation, fertilization management, and crop simulation to boost agricultural efficiency and sustainability. Reinforcement learning (RL), in particular, has outperformed traditional approaches in optimizing yields and managing resources. Yet, widespread AI adoption remains limited by discrepancies between algorithmic recommendations and farmers' practical experiences, local knowledge, and traditional practices. To bridge this gap, our study emphasizes Human-AI Interaction (HAII), specifically targeting transparency, usability, and trust in RL-driven farm management. We employ a well-established trust framework—consisting of ability, benevolence, and integrity—to construct a novel mathematical model quantifying farmers' confidence in AI-based fertilization strategies. Farmer surveys conducted specifically for this research highlight critical misalignments, and these insights are incorporated into our trust model, subsequently integrated into a multi-objective RL framework. Unlike previous methods, our approach directly embeds trust into policy optimization, ensuring AI-generated recommendations are technically robust, economically feasible, context-sensitive, and socially acceptable. By aligning technical performance with human-centered trust, this research provides a practical path toward broader AI adoption in agriculture.

1. Introduction

With the growing impact of climate change, farmers are struggling to maintain agricultural productivity as extreme weather events, such as heatwaves and droughts, disrupt field operations and threaten crop yields. Meanwhile, global food insecurity persists, with the Food and Agriculture Organization (FAO) reporting that nearly 828 million people suffered from hunger in 2022. To address these challenges, agriculture is increasingly adopting advanced technologies, including artificial intelligence (AI), aimed at improving efficiency and resilience. Innovations such as remote sensors for monitoring crops and soil conditions, large-scale drones for precision pest management, AI-powered systems for optimizing irrigation and fertilization, and advanced modeling and simulation tools are transforming farming practices. Together, these technologies define precision agriculture, a data-driven approach that promotes sustainable and efficient food production [1].

Recent advancements in AI-driven agricultural management have sparked significant research interest, with several studies offering valuable insights. Wu et al. [2] found that reinforcement learning (RL) techniques could surpass conventional methods in agricultural management generation, achieving similar or even better crop yields while significantly reducing fertilizer usage, marking a key step toward more sustainable practices. Similarly, Sun et al. [3] examined the use of RL for optimizing irrigation, demonstrating its ability to conserve water without negatively affecting crop health, and highlighted the usefulness of Gym-DSSAT, a crop simulation platform, in managing agricultural resources. Furthermore, Wang et al. [4] reinforced the effectiveness of RL-based fertilization strategies in challenging environments, further showcasing the potential of AI to transform modern farming.

While our earlier research showed that RL-generated agricultural management strategies could be successfully implemented in various climatic conditions [5], discussions with farmers have highlighted significant concerns. Although AI-driven policies (i.e., strategies) aim at maximizing economic outcomes, they often focus on theoretical optimal solutions rather than aligning with farmers' practical needs and experiences. This disconnect can undermine trust in AI-generated recommendations, resulting in their rejection or discontinuation in real-world farming practices.

The challenges mentioned earlier are closely related to the field of human-AI interaction (HAII), which examines how humans engage with

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https://doi.org/10.1016/j.atech.2026.102145
Received 15 May 2025; Received in revised form 14 April 2026; Accepted 21 April 2026
Available online 22 April 2026
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and respond to AI technologies. HAII has gained increasing importance in ensuring the successful adoption of AI. The field focuses on user-centered design, prioritizing aspects such as usability, transparency, interpretability, and responsiveness to human inputs and expectations [6]. Its goal is to create AI systems that are developed with a deep understanding of human behavior, values, and limitations, fostering more effective interactions between people and technology. Recent studies in HAII have shown that improving transparency and interpretability can significantly boost user trust, satisfaction, and long-term acceptance of AI-driven systems [7].

Trust models have a long-standing history, applied across a wide range of contexts, from personal relationships to intricate business dynamics. A variety of models have been developed to systematically define and measure trust in different scenarios. Early research on interpersonal trust laid the groundwork for understanding how trust develops and is maintained over time. For example, Lewicki and Bunker [14] introduced a sequential model of trust development, outlining stages of calculative, knowledge-based, and identification-based trust. Trust models have also been extensively used in organizational settings, where they emphasize the importance of fostering long-term cooperation and stable partnerships. One influential model in this area is Butler's Four-Component Model of Trust [15], which identifies integrity, competence, consistency, and openness as key factors influencing trust within organizations. This framework is particularly useful for assessing long-term business relationships, such as those between suppliers and clients.

As technology advances and digitalization increases, trust models have evolved to address trust in technology-mediated environments. Sectors such as e-commerce, digital finance, and virtual collaboration platforms have particularly benefited from these modified frameworks. More recently, the growing interaction between humans and AI systems has led to a surge in the development of specialized trust models. Ueno et al. [16] conducted an extensive review that categorized trust models into three types: cognitive, affective, and dispositional. Cognitive models focus on rational evaluations of AI performance and dependability, while affective models examine the emotional and relational aspects of trust. Dispositional models, on the other hand, investigate personal characteristics that influence individuals' perceptions of trustworthiness.

In agricultural communities, farmers have pointed to instances where AI-generated recommendations conflict with established practices or omit critical local context, including soil properties, historical pest pressures, and microclimate variability [8]. Prior surveys further suggest that many farmers perceive agricultural AI tools as "black boxes," lacking the transparency and interpretability needed to support appropriate trust and to evaluate long-term risks [9]. Together, these findings motivate the integration of human-centered human-AI interaction (HAII) principles into agricultural decision support, so that model-optimized strategies remain accountable to on-farm realities and user values [10]. Accordingly, we conduct a farmer-focused survey designed to surface recurring points of conflict, identify information needs for interpretation and validation, and elicit feedback on how AI recommendations could better align with practical constraints and decision routines.

These challenges are amplified as agricultural decision-support systems increasingly rely on advanced computational and optimization methods (e.g., bio- and nature-inspired algorithms), which can be difficult for end users to interpret and verify [11]. However, a key science gap remains: there is limited empirical and methodological work that translates farmer trust into measurable, actionable constructs grounded in real decision constraints [12]. In this context, improving adoption requires shifting emphasis from predictive or optimization performance alone toward human-in-the-loop validation processes: farmers must be able to interrogate recommendations, compare them against situational awareness (e.g., field history) and income-related constraints (e.g., expected profit impacts), and understand the assumptions under which a strategy is preferred. Designing transparent interfaces and two-way validation mechanisms is therefore essential for translating algorithmic outputs into recommendations that farmers can confidently evaluate and act upon [13].

To further address this crucial gap, we develop a trust model that assesses and quantifies farmers' confidence in AI-driven management strategies. Survey responses provide empirical grounding to refine the trust model so that it better reflects real-world perceptions and decision constraints, enabling agricultural management solutions that are more closely aligned with farmers' needs and more likely to achieve sustained adoption.

This research makes several key contributions, starting with the design of a comprehensive farmer survey that evaluates an expert fertilization plan alongside various AI-generated plans. The survey data analysis then informs the development of a quantitative trust model tailored to assess farmer trust in AI-driven agricultural recommendations. We adopt Mayer et al.'s [17] well-established three-dimensional framework, which comprises ability, integrity, and benevolence, as this trust model offers a more precise and practical assessment than earlier theoretical models [15,18]. By incorporating explicit data from farmers' real-world experiences, habits, expectations, and context-specific considerations, our trust model enhances its relevance to actual farming scenarios.

In addition, our study introduces a novel integration of the trust model into the RL optimization process through a multiple-objective setting. Unlike previous research [19], where trust was assessed only after policy development, our approach embeds trust evaluations within the RL training loop. This allows real-time feedback on trust during policy optimization, alongside traditional metrics like expected returns. By doing so, our approach ensures that AI-driven agricultural management strategies are not only economically optimal but also practically acceptable and aligned with farmers' expectations, thereby increasing acceptance, effectiveness, and long-term adoption of AI solutions in agriculture.

This paper is organized as follows. After the introduction, Section 2 presents the formulation of the partially observable Markov decision process (POMDP) and provides an overview of recurrent neural network (RNN)-based deep Q-learning, as well as multi-objective reinforcement learning (MORL). Section 3 outlines the simulation environment settings, defines the problem to be addressed, and generates various AI recommendations for fertilization based on different objectives, without incorporating trust. Section 4 designs and administers a survey for farmers to evaluate the AI-generated recommendations, analyzes the results, and develops the trust model utilized in this study. Section 5 integrates the trust model into the RL training process to further optimize the fertilization strategies and presents the simulation results. Finally, Section 6 concludes the paper with a summary of our findings, a discussion of implications, and suggestions for alternative solutions and directions for future research.

2. Methodology

2.1. POMDP

A Partially Observable Markov Decision Process (POMDP) extends the concept of Markov Decision Processes (MDPs) to accommodate decision-making under uncertainty due to partial observability of states. Unlike MDPs, where an agent can fully observe the current state, in a POMDP, the agent obtains indirect observations, thereby introducing inherent uncertainty regarding the true state of the environment. A POMDP can be represented by a tuple $\mathcal{P} = (S, s_0, A, T, O, \Omega, R)$, which includes the following components.

- A finite set of states: $S = \{s_1, \dots, s_n\}$.
- An initial state: $s_0 \in S$.
- A finite set of actions: $A = \{a_1, \dots, a_m\}$, where $A(s)$ denotes the set of actions available to the agent when in state s.
- A state transition probability function: $T : S \times A \times S \to [0, 1]$, where $T(s, a, s')$ represents the probability of transitioning from state s to

state s' after taking action a. This function satisfies the condition $\sum_{s' \in S} T(s, a, s') = 1$, ensuring that the total probability of transitioning to any possible next state sums to one.
- A finite set of observations: $O = \{o_1, \ldots, o_q\}$, where $O(s)$ represents the set of possible observations the agent can receive when it is in state s.
- An observation probability function: $\Omega : S \times A \times O \rightarrow [0, 1]$, where $\Omega(s', a, o)$ defines the probability of observing o after taking action a and arriving at state s'. This function also satisfies the condition $\sum_{o \in O} \Omega(s', a, o) = 1$, ensuring that the total probability of all possible observations at s' sums to one.
- A reward function: $R : S \times A \times S \rightarrow \mathbb{R}$, which assigns a numerical reward for transitioning from state s to state s' after taking action a. This function provides immediate feedback to guide the agent's learning and decision-making process.

The primary goal of an RL agent is to maximize the expected cumulative discounted reward (i.e., expected return or utility), which reflects the agent's objective of obtaining the highest possible long-term reward by making informed decisions at each time step [20]. Formally, this is expressed as:

$$U(s) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t, s_{t+1}) \middle| s_{t=0} = s\right] \tag{1}$$

where s_t represents the state of the environment at time t, while a_t denotes the action chosen by the agent at that time. The function $R(s_t, a_t, s_{t+1})$ specifies the immediate reward the agent receives for transitioning from state s_t to state s_{t+1} as a result of action a_t. The agent seeks to optimize its decision-making to maximize this cumulative reward. The parameter $\gamma \in [0, 1]$ is the discount factor, which is crucial for balancing immediate versus future rewards. A discount factor closer to 1 places greater emphasis on future rewards, making the agent prioritize long-term gains over short-term benefits. Conversely, a lower value of γ results in more emphasis on immediate rewards, with less concern for the long-term future. This balancing act between immediate and future rewards is central to decision-making, as agents must weigh the trade-offs and make strategic choices.

In the context of partial observability, the agent does not have direct access to the exact state s_t at each time step. Instead, it relies on observations, which may be incomplete, noisy, or ambiguous, to form beliefs about the underlying state of the system. This uncertainty in the agent's perception of the environment significantly complicates the decision-making process, as the agent must infer the most likely state from its observations rather than directly observing the state [21]. As a result, the agent's decision-making is not solely focused on optimizing for immediate rewards; it must also account for the uncertainty inherent in its belief about the state. These beliefs, typically represented as a probability distribution over all possible states, guide the agent's action by providing a way to quantify uncertainty and make informed decisions based on its current knowledge [22].

To address this challenge, the agent typically maintains a belief state, which serves as a representation of its uncertainty about the system's true state. This belief state is a probability distribution over all possible states, and it evolves as the agent gathers more observations. Through a process known as belief updating, the agent refines its belief over time, which in turn influences the selection of future actions [23]. The evolving belief allows the agent to make more informed decisions but also introduces the challenge of managing this uncertainty over time. Consequently, the agent's ability to make optimal decisions is shaped not just by the immediate effects of its actions but by the long-term consequences, which depend on how its belief about the state evolves.

One way to handle partial observability in RL is to transform the POMDP problem into a corresponding MDP one in the belief space [24]. This belief space is a probability distribution over states, representing the agent's uncertainty about the environment. In this formulation, the agent's objective is to find an optimal policy in the belief space, rather than directly in the state space. This approach allows the agent to handle uncertainty by making decisions based on its belief about the state, rather than a precise state observation [25]. However, this transformation can be computationally demanding and may not always be feasible when transition and observation probabilities are unknown.

Many model-based methods for solving partially observable problems rely on knowledge of the transition and observation probabilities, which are often difficult to obtain in real-world scenarios. Model-free methods, by contrast, do not require explicit knowledge of the environment's dynamics; instead, they learn from interactions with the environment and update their policies based on observed rewards and outcomes [26]. These methods are particularly useful in situations where the transition and observation probabilities are uncertain or too complex to model directly, making them a suitable approach for solving real-world POMDP problems.

From a technical perspective, formulating the agricultural management problem as a POMDP rather than a standard MDP is critical for bridging the gap between simulated training environments and real-world deployment. A standard MDP assumes full observability of the environment's true state. In agricultural systems, achieving full observability would require continuous and precise measurements of complex subsurface variables, such as deep soil nitrogen levels, detailed root-zone moisture profiles, and dynamic crop biomass. In practice, obtaining this complete state representation is prohibitively expensive and logistically impractical, as it would require dense sensor networks or frequent laboratory-based measurements. The POMDP framework naturally accommodates these operational constraints by explicitly modeling uncertainty in the system state. With a limited set of practically available observations (e.g., surface weather data and periodic standard soil tests), the decision-making agent can still produce robust management policies. Furthermore, as demonstrated in our previous work [4], POMDP-based agents can achieve agronomic performance, measured in crop yield and resource-use efficiency, comparable to that obtained under a full-state MDP formulation, while relying only on a substantially reduced and practically observable set of inputs.

2.2. RNN-based DQN

Q-learning [27] is a widely-used model-free RL method that leverages Q-values, also referred to as action values or state-action values, to evaluate and guide action selection throughout the learning process. The Q-value, denoted as $Q(s, a)$, represents the expected cumulative reward an agent can obtain by taking action a in state s and subsequently following a given policy. Through iterative updates based on observed rewards and transitions, Q-learning enables an agent to learn an optimal policy without requiring prior knowledge of the environment dynamics. Traditional tabular Q-learning methods are effective for environments with finite and discrete state spaces, where Q-values can be explicitly stored and updated in a table. However, in complex real-world applications, such as agricultural management, where state and action spaces are often continuous or extremely large, tabular methods become impractical due to their inability to generalize across states efficiently.

To address this limitation, function approximation techniques, such as deep Q-networks (DQNs) and other neural network-based methods, have been developed to estimate Q-values in a more scalable and efficient manner, enabling RL to be applied to complex problems. DQN [28] leverages deep neural networks (DNNs) to approximate Q-values, allowing the method to generalize across vast or continuous state-action spaces where traditional tabular approaches fail. A key feature of DN is the use of two neural networks with identical architectures: an evaluation Q-network Q_e and a target Q-network Q_t.

The evaluation network is responsible for generating Q-value estimates and is continuously updated during training, while the target network provides stable Q-value targets for learning. To prevent instability and divergence issues caused by highly correlated updates, the target network's weights are periodically synchronized with those of the evaluation network rather than updated at every step. This mechanism improves convergence and helps mitigate oscillations in Q-value estimates. Additionally, DQN employs experience replay, a technique that stores past experiences in a buffer and randomly samples minibatches for training. This process breaks the correlation between consecutive experiences, further stabilizing learning and enhancing sample efficiency. These improvements make DQN well-suited for solving high-dimensional RL problems.

In real-world applications, observations provided to an agent often contain partial information about the underlying environment state, making these scenarios better modeled as POMDPs rather than fully observable MDPs. Decision-making in POMDPs necessitates the agent to integrate information over time, relying on sequences of past observations rather than single, isolated observations to infer the hidden state of the environment. To address this challenge and effectively capture temporal dependencies in observation sequences, we introduced a Recurrent Neural Network (RNN) into the DQN architecture [29]. Specifically, the gated recurrent unit (GRU) [30] is adopted in this study, so the Q-network can retain and utilize relevant historical observations, enabling the agent to effectively model long-term dependencies in partially observable environments, reducing the ambiguity caused by incomplete observations [31].

In this framework, our DQN incorporates two GRU-based Q-networks: an evaluation network and a target network. These networks are represented as $Q_E(\mathbf{o}_t, a_t; \theta_E)$ and $Q_T(\mathbf{o}_t, a_t; \theta_T)$, respectively, where θ_E and θ_T denote their corresponding sets of trainable parameters. At each discrete time step t, the agent observes a partial representation of the environment. It then generates an observation sequence that includes a short history of past observations, allowing the recurrent network to capture temporal dependencies and infer hidden state information. This sequence is denoted as $\mathbf{o}_t$. Based on the Q-values predicted by the evaluation network, the agent selects an action a_t. The selection process follows an ϵ-greedy technique, which balances exploration and exploitation: with probability ϵ, the agent explores by selecting a random action, and with probability 1-ϵ, it exploits by choosing the action with the highest predicted Q-value. This technique ensures that the agent continues discovering new strategies while prioritizing high-reward actions.

Once the agent executes the selected action a_t, the environment transitions to a new state, generating a new observation, and providing a scalar reward r_t that reflects the immediate benefits of the action. The observation sequence is updated by incorporating the new observation into $\mathbf{o}_{t+1}$, and the agent stores the experience as a tuple $\left(\mathbf{o}_t, a_t, r_t, \mathbf{o}_{t+1}\right)$ in an experience replay memory [37]. This memory buffer plays a crucial role in stabilizing training by allowing the evaluation network to learn from a diverse set of past experiences rather than sequentially correlated transitions, which can introduce bias and hinder convergence. Specifically, the evaluation Q-network is updated using the Bellman equation given in Eq. (2):

$$Q_{\text{new}}(\mathbf{o}_t, a_t) = Q_E(\mathbf{o}_t, a_t; \theta_E) + \alpha\left[r_t + \gamma \max_{a' \in A} Q_t(\mathbf{o}_{t+1}, a_{t+1}; \theta_T) - Q_E(\mathbf{o}_t, a_t; \theta_E)\right] \tag{2}$$

where α is the learning rate. The term $\max_{a' \in A} Q_t(\mathbf{o}_{t+1}, a_{t+1}; \theta_T)$ represents the highest Q-value predicted by the target network for the update observation sequence $\mathbf{o}_{t+1}$, ensuring that the agent accounts for long-term rewards when making decisions.

During training, mini-batches of experience tuples are randomly sampled from the replay buffer to update the evaluation network. This batch-based learning strategy helps break the correlation between consecutive experiences, leading to more robust learning dynamics. Meanwhile, to maintain training stability, the target network parameters θ_T are not updated continuously but rather copied periodically from the evaluation network. This delayed update mechanism prevents oscillations in Q-value targets and reduces the risk of divergence. Through the combination of recurrent network architectures, experience replay, and separate Q-networks, our approach effectively captures temporal dependencies in sequential decision-making tasks while ensuring stable and efficient learning.

2.3. MORL

In RL, the traditional framework typically uses a single scalar reward function, $R(s, a, s')$, to guide the agent's decision-making process. However, real-world applications often involve multiple, potentially conflicting objectives that are not easily captured by a single reward. To address this complexity, our study employs Multi-Objective Reinforcement Learning (MORL) within the framework of POMDPs. Unlike traditional RL, MORL extends the single-objective reward function to a multi-objective reward vector, $\vec{R}(s, a, s')$, which enables the agent to consider several competing objectives simultaneously [32].

This extension is formalized as:

$$\vec{R}(s, a, s') = \left(R_1(s, a, s'), R_2(s, a, s'), \ldots, R_k(s, a, s')\right) \tag{3}$$

where each component $R_i(s, a, s')$ corresponds to a distinct objective, such as performance, efficiency, or safety. In this framework, the agent's goal shifts from optimizing a single scalar quantity to simultaneously optimizing a vector of objectives, each of which may have its own distinct rewards associated with different state-action transitions. This more complex reward structure reflects the real-world scenario where multiple factors must be balanced, often in the presence of trade-offs and conflicts, making more informed and adaptable decisions in dynamic environments.

Due to the multiplicity and potential conflict among objectives, optimality within MORL is defined through the concept of Pareto dominance rather than scalar maximization employed in standard RL. A policy is considered Pareto optimal if no other policy exists that can improve all objectives simultaneously without causing at least one objective to deteriorate [33]. The set of all such Pareto-optimal policies forms what is known as the Pareto front, representing the spectrum of optimal trade-offs among competing objectives.

The Pareto front serves as a critical tool for decision-making in MORL, offering a structured view of the diverse solutions available. Instead of relying on a single, predefined objective function, decision-makers select policies aligned with dynamic preferences and specific contextual constraints [34]. This flexibility is particularly valuable in complex, real-world scenarios such as autonomous driving, medical decision-making, and robotic control, where trade-offs must be carefully managed. By leveraging the Pareto front, MORL enables adaptive and informed decision-making, ensuring that chosen policies reflect the most appropriate balance between competing goals.

In practice, there are multiple strategies for choosing a final policy from the Pareto front. One common approach is a posteriori selection: after learning, the entire set of Pareto-optimal solutions is presented to a decision-maker, who then picks the most suitable policy based on domain knowledge or situational requirements. Alternatively, if user preferences are known or can be elicited at deployment time, scalarization methods (e.g., linear weighted sums, Chebyshev metrics, or other utility-based measures) can be applied to evaluate each Pareto solution according to the current preference, and the policy with the best scalar score is selected [35]. This combination, preserving the Pareto structure during learning, then applying preference-based filtering or ranking at selection, retains the flexibility of the multi-objective approach, while enabling decision-makers to adaptively choose among potentially conflicting objectives.

The extension of the Q-learning algorithm to MORL contexts involves modifying the conventional scalar Bellman update rule to incorporate vector-valued rewards and dynamic preferences. To approximate the entire Pareto front efficiently, we employ a Conditioned Network approach, where the value function is conditioned on a preference vector $\mathbf{w}$ representing the relative importance of objectives [36]. This allows the agent to learn a family of optimal policies simultaneously within a single model. The modified MORL update rule is formally expressed as:

$$\vec{Q}_{\text{new}}(\mathbf{o}, a, \mathbf{w}) = \vec{Q}(\mathbf{o}, a, \mathbf{w}) + \alpha\left[\vec{R}(s, a, s') + \gamma\vec{Q}(\mathbf{o}', a^*, \mathbf{w}) - \vec{Q}(\mathbf{o}, a, \mathbf{w})\right] \quad (4)$$

where $\vec{Q}(\mathbf{o}, a, \mathbf{w})$ denotes the vector-valued Q-function that estimates the expected discounted cumulative return for each objective. The term a^* represents the optimal action at the next observation $\mathbf{o}'$ with respect to the current preference $\mathbf{w}$, defined as:

$$a^* = \underset{a' \in A}{\operatorname{argmax}}\left(\mathbf{w} \cdot \vec{Q}(\mathbf{o}', a', \mathbf{w})\right) \quad (5)$$

This formulation replaces the standard scalar maximization with a scalarized maximization over the vector outputs. Unlike single-objective Q-learning, which converges to a single optimal policy, this preference-conditioned update rule enables the agent to generalize across the weight space. During the learning process, the preference $\mathbf{w}$ is sampled systematically to ensure the agent explores various trade-offs on the Pareto front.

By explicitly conditioning the policy on $\mathbf{w}$, the algorithm avoids the computational intractability of maintaining explicit Pareto sets for every state. Instead, the Pareto front is implicitly represented by the continuous manifold of policies learned by the network. At deployment time, practitioners can retrieve a specific Pareto-optimal policy simply by querying the network with the desired preference vector $\mathbf{w}$, allowing for instant adaptation to changing requirements without re-training.

3. Intelligent agricultural management

3.1. Agricultural environment and reinforcement learning

In this research, we utilized Gym-DSSAT [38], an advanced virtual simulation platform tailored to model crop growth, yield performance, and environmental effects like nitrate leaching. This simulation takes into account different weather conditions and initial soil parameters, offering a reliable framework for evaluating agricultural practices and management approaches. Gym-DSSAT exposes 28 variables that constitute the observation vector provided to the agent. These include endogenous soil and crop physiological variables (e.g., soil moisture and crop growth stage), as well as exogenous climate forcing variables such as temperature and precipitation. This comprehensive set of variables facilitates precise and consistent agricultural modeling.

It is important to note that all figures, tables, and responses presented in Sections 3 and 4 are reported in U.S. customary units. The figures and tables in these sections reproduce the exact materials shown to participants as part of the survey instrument. Because all participants were U.S. farmers, we report all quantities exactly as they were presented to and evaluated by the respondents in order to preserve the fidelity of the survey and accurately reflect their interpretations and feedback. The use of U.S. customary units in these sections is therefore intentional and context-specific. In contrast, simulation results and analytical outcomes reported elsewhere in the manuscript are expressed in SI units.

However, as highlighted in our previous work [4,39], agricultural systems are inherently complex and subject to partial observability due to the wide range of interacting factors, including variable weather patterns, soil diversity, and plant behavior. To manage the uncertainty that comes with these variables, we selected ten essential observations as the core observational inputs for the decision-making agent. These variables, outlined in Table 1, were carefully chosen for their relevance to real-world agricultural monitoring and their ability to be measured practically, ensuring that the insights from our simulation can be effectively applied to actual farming operations.

The interaction between the RL agent and the agricultural environment, modeled using the DSSAT simulator, is illustrated in Fig. 1. In this setup, the RL agent continuously interacts with the simulated environment to learn optimal agricultural management strategies tailored to specific weather patterns. Given that corn cultivation in Iowa predominantly relies on natural rainfall rather than irrigation [4], we intentionally excluded irrigation practices from our analysis. Instead, we focused exclusively on optimizing nitrogen fertilizer application strategy, a critical factor influencing crop health, productivity, and environmental impact. Proper nitrogen management is essential for balancing crop nutrient requirements while mitigating potential issues like nitrate leaching.

Table 1
Observations of the agricultural environment used in this study.

cumsumfert	cumulative nitrogen fertilizer applications
dap	days after planting
istage	DSSAT maize growing stage
pltpop	plant population density
rain	rainfall for current day
sw	volumetric soil water content in soil layers
tmax	maximum temperature for the current day
tmin	minimum temperature for the current day
vstage	vegetative growth stage (number of leaves)
xlai	plant population leaf area index

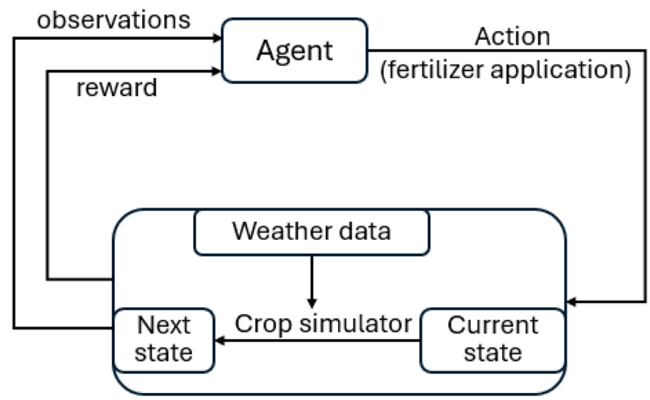

Fig. 1. The interaction between an RL agent and the agricultural environment.

To enable the RL agent to explore and learn effective fertilizer application strategies, we defined its action space as a range of nitrogen application rates per day. These rates are discretized into increments of 5 kg/ha, ensuring a balance between granularity and computational feasibility. Mathematically, this results in a discrete action variable, k, which takes integer values from 0 to 40. Each value corresponds to a specific daily nitrogen application rate, ranging from 0 kg/ha (no nitrogen applied) to a maximum of 200 kg/ha. By systematically evaluating different nitrogen application strategies, the RL agent aims to identify policies that enhance crop yield while optimizing resource efficiency and reducing environmental impact.

During the learning process, the agent refines its decision-making by receiving environmental feedback on its actions. This iterative process enables the agent to develop optimal nitrogen management strategies that maximize crop yield, minimize resource waste, and promote sustainable farming practices. The agent's decisions are guided by a carefully designed reward function, formulated to capture the economic, agronomic, and environmental trade-offs associated with nitrogen application. The reward function is expressed in Eq. (6), incorporating key factors such as economic profitability, fertilizer efficiency, and environmental sustainability:

$$R_t = \begin{cases} w_1 Y - w_2 N_t - w_3 L_t - w_4 N_F & \text{at harvest} \\ -w_2 N_t - w_3 L_t & \text{otherwise} \end{cases} \quad (6)$$

Here N_t represents the nitrogen application on each simulation day t, while L_t (kg/ha) denotes the corresponding nitrate leaching, a critical environmental impact factor calculated by the DSSAT simulator. The reward function differentiates between daily operations and the fi-

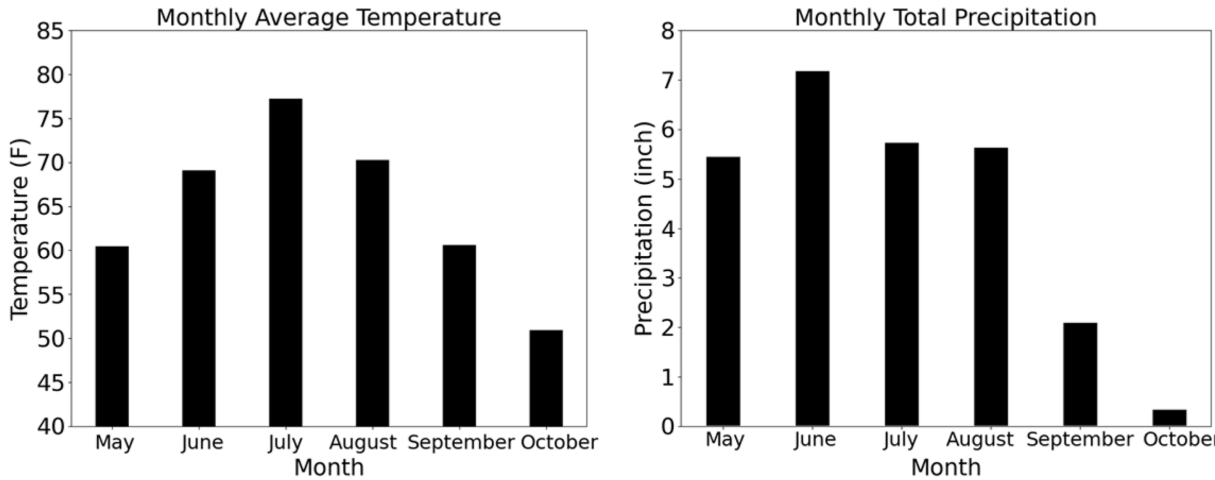

Fig. 2. Average monthly temperature and total precipitation during the corn growth period in Ames, Iowa, in 1999.

nal harvest period. During regular growth phases, the reward penalizes excessive nitrogen use and nitrate leaching. At harvest, the reward additionally accounts for crop yield Y (kg/ha), reinforcing the economic benefits of improved agricultural productivity. Moreover, the term N_F captures the total number of fertilizer applications required throughout the growing season, introducing an implicit cost associated with labor and operational efforts.

The weighting factors $w_i, i = 1 \ldots 4$ are carefully chosen to reflect real-world economic and environmental considerations. Specifically, $w_1 = 0.22$ represents the market price of corn per kilogram in 2023, while $w_2 = 1.5$ corresponds to the price of nitrogen per kilogram. The coefficient $w_3 = 15$ assigns a penalty for nitrate leaching, set as ten times w_2 [4], emphasizing its significant environmental consequences. Finally, $w_4 = 3$ accounts for labor costs per fertilizer application, recognizing the operational burden associated with frequent nitrogen applications.

This reward formulation ensures that the RL agent is incentivized to adopt efficient nitrogen management practices that optimize yield while mitigating excess fertilizer use and environmental degradation. By incorporating economic and ecological factors, the model fosters a balanced approach to sustainable corn production. It should be noted that w_3 or w_4 may be set to zero depending on specific objective considerations, as discussed in the subsequent section. The problem of intelligent agricultural management is therefore defined below.

Problem 1. A POMDP, as defined in Section 2.1, models the agricultural nitrogen fertilizer management task, where the environmental dynamics are simulated using the Gym-DSSAT framework. The reward function, defined in Eq. (6), is formulated as a weighted sum of multiple components, including economic profit (yield), nitrogen fertilizer usage, nitrate leaching penalties, and labor costs associated with fertilizer applications. These weighting factors allow for flexibility in reflecting diverse objectives; for instance, certain scenarios may prioritize yield maximization while disregarding labor cost or environmental impact. The objective is to derive an optimal policy that generates fertilization management strategies to maximize the cumulative reward, wherein the reward structure can be tailored to align with specific economic priorities or environmental considerations.

3.2. AI-generated recommendations

In this study, AI-generated fertilization recommendations are derived from optimal policies obtained through reinforcement learning, as described in Section 3.1. The selected study site is a field located at the Agronomy and Agricultural Farm of Iowa State University in Ames, Iowa (42.020°N, 93.750°W). To ensure realistic and representative outcomes, we used historical weather data from Iowa in 1999, a year without extreme weather events. The average monthly temperature and total monthly precipitation during the corn growth period are illustrated in Fig. 2, providing essential environmental context for the simulation. Both the soil characteristics and historical weather data were obtained from DSSAT.

In all scenarios, the agent aims to develop optimal nitrogen management strategies that maximize crop yield while minimizing fertilizer waste. We consider four distinct scenarios that introduce additional objectives by adjusting the weighting factors w_i $(i = 1 \ldots 4)$ in Eq. (6).

- Scenario 1 integrates environmental goals by penalizing nitrate leaching (w_3) alongside yield and fertilizer costs, but disregards labor costs ($w_4 = 0$)
- Scenario 2 focuses solely on the trade-off between crop yield and fertilizer input, omitting both nitrate leaching and labor cost considerations ($w_3 = w_4 = 0$)
- Scenario 3 applies the comprehensive weighting scheme described in Eq. (6), accounting for yield, fertilizer use, nitrate leaching, and labor costs simultaneously.
- Scenario 4 incorporates labor costs (w_4) into the economic optimization but neglects environmental impacts by setting $w_3 = 0$.

To learn the optimal policy for each scenario, we employ the ϵ-greedy selection strategy to balance exploration and exploitation in RL. A discount factor of 0.99 is used to emphasize the importance of future rewards in the decision-making process. The design and update of neural networks, as Q-networks, are implemented using PyTorch, with the Adam optimizer configured with an initial learning rate of 3×10^{-5} and a batch size of 640. These hyperparameters are selected to strike a balance between model performance and computational efficiency. All simulations are performed on a system equipped with an Intel Core i7-12700K processor, an NVIDIA GeForce RTX 3070 Ti graphics card, and 64GB of RAM.

Once the optimal policies are learned, the corresponding fertilization recommendations - designated AI Recom 1, AI Recom 2, AI Recom 3, and AI Recom 4 - are derived and illustrated in Fig. 3. For benchmarking purposes, we additionally include the expert recommendation (Exp Recom) provided in Gym-DSSAT, which represents the fertilizer application policy used in the original Gym-DSSAT experiments and was developed by agronomic experts. Each plot in Fig. 3 shows the timing and amount of nitrogen fertilizer applications, along with the total quantity applied and the number of application events.

Additionally, the agricultural outcomes associated with all five fertilization recommendations are summarized in Table 2. The net income, while calculated slightly differently from the reward function defined

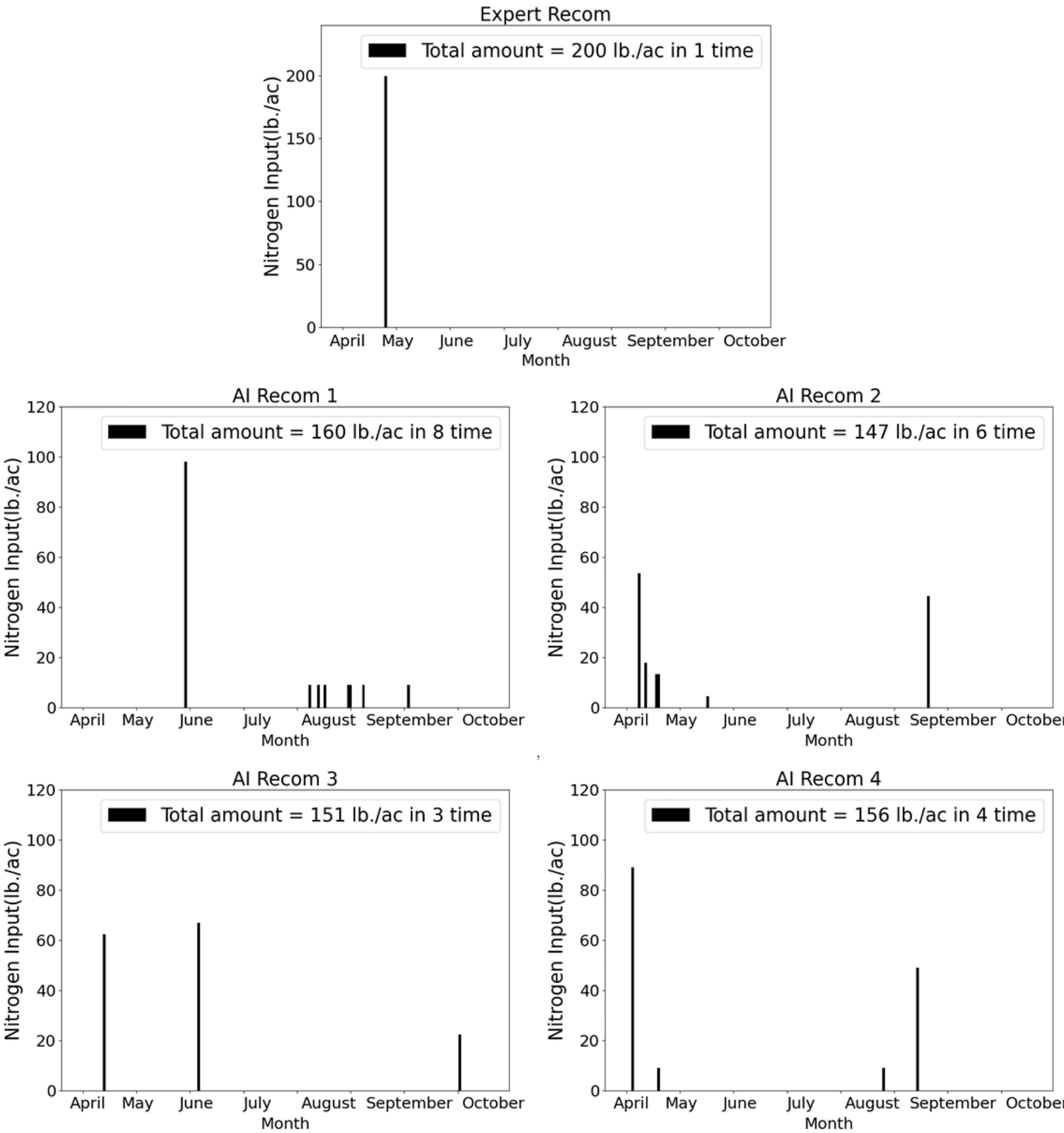

Fig. 3. Five fertilization recommendations generated under different policy scenarios.

in Eq. (6), is determined by subtracting the costs of fertilizer and labor from the market value of the crop yield, without accounting for environmental impacts. The expert recommendation serves as the baseline, with 100% environmental impact corresponding to the resulting nitrate leaching. Environmental impacts from other recommendations are then quantified relative to this baseline, based on their respective levels of nitrate leaching. As a result, the AI-generated recommendations demonstrate significant reductions in environmental impact, most notably AI Recom 1 and AI Recom 3, with impacts of only 35% and 38%, respectively.

Furthermore, while all recommendations achieve similar yield, AI Recom 1 results in the highest, albeit at the expense of increased labor costs due to eight fertilizer applications. AI Recom 3, meanwhile, offers an optimal balance among yield, labor cost, fertilizer usage, and environmental sustainability. It achieves slightly less yield than the expert recommendation while incurring significantly lower environmental impact and labor costs. This balance is reflected in its highest net income of $708 per acre.

4. Farmer survey and trust model

4.1. Farmer survey

In this study, we aim to understand the factors that influence farmers' preferences and trust in agricultural management strategies. We collected data through a survey and used the insights gained to develop a novel trust model. The survey was conducted in the Midwest region of the United States. Its specific objective is to gather feedback on various fertilization management strategies recommended by either human experts or AI systems. The survey assumes normal weather conditions, with the average monthly temperature and total monthly precipitation presented in Fig. 2. The focus is on nitrogen fertilizer management, par-

Table 2
Outcomes of each fertilization management recommendation.

	Corn yield (bu/ac)	Number of fertilization	Fertilizer amount (lb/ac)	labor cost ($)	Impact to environment	Net income ($/ac)
Exp Recom	147	1	200	3	100%	684
AI Recom 1	147.46	8	160	24	35%	693
AI Recom 2	146.15	6	147	18	62%	700
AI Recom 3	146.42	3	151	9	38%	708
AI Recom 4	147.39	4	156	12	54%	707

ticularly the timing and quantity of applications throughout the corn growth cycle.

First, we present participants with five fertilization recommendations illustrated in Fig. 3. Participants are initially asked to evaluate these recommendations using a 5-point Likert scale. Specifically, they rate their preference from 1 (least preferred) to 5 (most preferred) without knowing the source—whether generated by a human expert or an AI system. In addition to preference, participants also rate their trust in each recommendation on a corresponding 5-point Likert scale ranging from 1 (extremely unlikely to trust) to 5 (extremely likely to trust).

Next, we provide participants with detailed explanations of all five recommendations, as outlined in Section 3.2, along with their corresponding outcomes listed in Table 2. Specifically, we clarify that nitrate leaching from the expert-generated recommendation serves as the baseline for measuring the environmental impact of the AI-generated recommendations. Net income is calculated based on corn revenue, fertilizer costs, and labor expenses. In addition to maximizing corn yield and minimizing fertilizer use, the AI-generated recommendations are optimized according to varying trade-offs between nitrate leaching — which affects environmental impact — and labor costs. After reviewing these detailed explanations, participants re-rank their preferences for the recommendations and reassess their trust ratings.

Following this reassessment, the survey asks participants to assign weights to four key decision-making factors - corn yield, fertilizer amount, fertilization frequency, and environmental impacts - ensuring that the total weight sums to 100%. Finally, participants respond to two questions related to their farming practices: “For seasonal corn management, how many nitrogen fertilization applications do you believe are most appropriate?” and “How much nitrogen fertilizer do you typically apply per acre before the corn is harvested?”

4.2. Survey data and analysis

A total of 71 farmers participated in the survey, with Iowa having the highest proportion of respondents. After excluding 4 incomplete surveys and 1 from outside the U.S., 66 complete surveys remained. Further quality checks removed problematic responses — such as duplicate rankings or uniform answers - resulting in 54 usable surveys. The final sample comprised 33 males, 19 females, 1 non-binary respondent, and 1 who preferred not to disclose their gender, with an average age of 36.9 years.

Tables 3 and 4 present participants’ preference ranking before and after they were informed about the source of each recommendation - whether it was expert- or AI-generated - as well as the associated outcomes. The overall ranking was calculated using a weighted average, where the percentage of responses at each ranking level served as the weight. For example, prior to receiving the explanation, 20.4% of respondents ranked the expert recommendation as the most preferred, while 11.1% ranked it as the least preferred, with the remaining rankings distributed as shown in Table 3. The overall ranking score was computed as $0.204 * 5 + 0.185 * 4 + 0.259 * 3 + 0.241 * 2 + 0.111 * 1 = 3.13$.

Initially, AI Recommendation 4 received the highest preference score (3.21), followed closely by the expert recommendation (3.13), AI Recommendation 1 (3.11), and AI Recommendation 3 (3.09). AI recommendation 2 was the least preferred, with a notably lower score of 2.47. After participants were provided with explanations, preferences shifted

Table 3
Participants’ preferences for each recommendation before receiving an explanation, rated on a scale from 1 (least preferred) to 5 (most preferred).

	1	2	3	4	5	overall
Exp Recom	11.1%	24.1%	25.9%	18.5%	20.4%	3.13
AI Recom 1	20.4%	9.3%	29.6%	20.4%	20.4%	3.11
AI Recom 2	31.5%	25.9%	20.4%	9.3%	13.0%	2.47
AI Recom 3	9.3%	33.3%	11.1%	31.5%	14.8%	3.09
AI Recom 4	27.8%	7.4%	13.0%	20.4%	31.5%	3.21

Table 4
Participants’ preferences for each recommendation after receiving an explanation.

	1	2	3	4	5	overall
Exp Recom	16.7%	37.0%	9.3%	13.0%	24.1%	2.91
AI Recom 1	31.5%	20.4%	16.7%	20.4%	11.1%	2.60
AI Recom 2	20.4%	11.1%	35.2%	18.5%	14.8%	2.96
AI Recom 3	18.5%	20.4%	25.9%	29.6%	5.6%	2.83
AI Recom 4	13.0%	11.1%	13.0%	18.5%	44.4%	3.70

significantly. AI Recommendation 4 retained its top rank, but its preference score (3.70) became substantially higher than those of the other options. It was followed by AI Recommendation 2, the expert recommendation, AI Recommendation 3, and finally AI Recommendation 1, as shown in Table 4.

These findings suggest that farmers initially favored AI Recommendation 4 and the expert recommendation due to their familiarity and alignment with established farming practices. After receiving detailed explanations, the strong preference of AI Recommendation 4 persisted, indicating broad recognition of its practical advantages. In contrast, AI Recommendations 1 and 3, both of which explicitly incorporated environmental considerations, were less favorably received.

Tables 5 and 6 illustrate participants’ trust in each recommendation before and after receiving detailed explanations. Higher scores in these tables indicate greater trust. Initially, the expert recommendation received a moderate trust score of 3.42, slightly trailing AI Recommendation 2, which had the highest initial trust level at 3.52. AI Recommendations 3, 4, and 1 followed with lower trust scores of 3.30, 3.21, and 3.04, respectively. However, after the explanations were provided, the expert recommendation experienced the largest increase in trust, reaching the highest trust score of 3.67. Trust in AI Recommendations 3 and 4 also rose substantially, to 3.58 and 3.39, respectively. In contrast, AI Recommendation 1 saw only a modest increase (to 3.13), while trust in AI Recommendation 2 slightly declined to 3.39, aligning more closely with its preference ranking.

Interestingly, participant preferences did not consistently align with trust ratings. Preferences were somewhat more flexible, whereas trust appeared more stable. The divergence between trust in a recommendation and an individual’s ultimate preference can be understood by considering the distinct foundations of these two constructs. Trust is often grounded in a cognitive assessment of the recommendation’s source. We deem a recommender trustworthy based on perceptions of its competence, benevolence, and integrity [17].

Table 5
Participants' trust for each recommendation before receiving an explanation, rated on a scale from 1 (extremely unlikely to trust) to 5 (extremely likely to trust).

	1	2	3	4	5	overall
Exp Recom	5.6%	24.1%	18.5%	25.9%	25.9%	3.42
AI Recom 1	14.8%	16.7%	31.5%	24.1%	13.0%	3.04
AI Recom 2	5.6%	13.0%	22.2%	42.6%	16.7%	3.52
AI Recom 3	3.7%	24.1%	27.8%	27.8%	16.7%	3.30
AI Recom 4	9.3%	24.1%	20.4%	29.6%	16.7%	3.21

Table 6
Participants' trust for each recommendation after receiving an explanation.

	1	2	3	4	5	overall
Exp Recom	9.3%	13.0%	16.7%	24.1%	37.0%	3.67
AI Recom 1	7.4%	20.4%	35.2%	25.9%	11.1%	3.13
AI Recom 2	7.4%	9.3%	40.7%	22.2%	20.4%	3.39
AI Recom 3	1.9%	14.8%	27.8%	35.2%	20.4%	3.58
AI Recom 4	3.7%	14.8%	33.3%	35.2%	13.0%	3.39

However, a gap often exists between an individual's level of trust and their expressed preferences. This gap arises partly because preferences are not fixed entities waiting to be discovered; rather, they are frequently constructed at the moment of decision and are highly malleable, shaped by context, choice framing, available options, and transient personal goals [40,41]. For example, a trusted financial advisor might recommend a diversified, long-term growth fund (a suggestion reflecting competence and benevolence), but an individual's preference at that moment might be swayed by a news headline about a speculative tech stock (framing and context) or a sudden short-term financial need (current goals).

Furthermore, while trust is often based on logical assessment, preferences, particularly for experiential choices like entertainment, food, or travel, are strongly affective in nature. They hinge on how an option makes us feel [42]. Someone may trust a critic's expertise yet prefer a lighthearted comedy for emotional lift. In such cases, a cognitively trusted recommendation may fail to resonate on the affective level, creating a disconnect with momentary preference.

We also believe the small sample size contributed to this discrepancy. Individual variations, particularly strong anti-AI sentiments among a few participants, may have disproportionately influenced overall trends. Additionally, some participants exhibited a deep-rooted trust in human expertise, suggesting enduring attitudes rather than situational preferences. Moreover, the limited sample may not adequately capture the diversity within the farming community. Factors such as age, education, farming experience, and technology acceptance significantly influence perceptions and trust in decision-support tools. Underrepresentation of specific subgroups could thus skew overall trust assessments between AI and expert recommendations.

The justifications provided by participants suggest that some farmers prioritized the timing and frequency of fertilization. Many noted that their trust in a recommendation was primarily driven by its alignment with their existing fertilization schedule. For example, some mentioned that they were more likely to trust and adopt a strategy if it scheduled fertilizer applications in April or May, as this aligned with their traditional practices (e.g., "I think April or May is the best combination of rainfall and growing plants taking advantage of the nitrogen"). Furthermore, participants indicated that the most significant factor influencing changes in their preference was net income, which was only disclosed after the explanations were provided. Many participants explicitly stated that net income was their primary consideration (e.g., "Net income is what matters most to me").

Among the four decision-making factors presented in the survey - corn yield, fertilization amount, fertilization frequency, and environmental impact - farmers prioritized corn yield the highest at 40.24%, followed by fertilization amount at 23.91%. Fertilization frequency ranked third at 19.02% while environmental impact was the least prioritized, with 16.83% of participants mentioning it.

Additionally, our survey included two general questions: "For seasonal corn management, how many nitrogen fertilization applications do you believe are most appropriate?" and "How much nitrogen fertilizer do you typically apply per acre before the corn is harvested?" Among all valid responses, about three-fourths of participants indicated that the ideal fertilization frequency is two or three applications per growth cycle. The most commonly cited fertilizer amounts were 150 lb/acre and 200 lb/acre, with other responses falling within this range.

4.3. Trust model

Trust in AI within agriculture extends well beyond decision support systems for fertilization or irrigation. It encompasses a wide variety of applications, including autonomous machinery, yield prediction, disease detection, and supply chain optimization. Across all these domains, trust shapes adoption decisions, usage patterns, and long-term reliance. Importantly, the identity of the "trustee" can vary across application settings: in some cases, it may be perceived as the AI system, whereas in others it may be associated with the organization deploying the system or the broader socio-technical arrangement surrounding its use. In the present study, we focus on a decision-support setting in which the trustee is the AI-based fertilization recommendation system as perceived by farmers as end users.

Trust is a multifaceted social and psychological phenomenon shaped by various factors, including social networks, personal charisma and appearance, confidence levels, perceived competence in task execution, and credibility established through past interactions and connections [46]. It serves as the cornerstone of successful relationships between humans and non-human agents, e.g., AI agents. When these relationships involve dependence and risk - especially given the complexity and non-deterministic behavior of AI systems - trust becomes crucial. Misplaced or insufficient trust can result in misuse, overreliance, or outright rejection of the technology. Moreover, the successful integration of AI into workplace processes depends on workers' confidence in its capabilities and reliability [47].

Trust has been defined in multiple ways across disciplines. Mayer, Davis, and Schoorman [17] conceptualize trust as the willingness of a party to be vulnerable to the actions of another party based on the expectation that the latter will perform actions important to the trustor. In contrast, research in human-AI interaction often adopts the definition proposed by Lee and See [43], and referenced by Ueno et al. [16], which frames trust as "the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability." Although both definitions involve reliance under uncertainty, Mayer et al. foreground willingness to accept vulnerability and risk, whereas Lee and See emphasize perceived goal facilitation under such conditions.

We adopt Mayer et al.'s [17] definition for two main reasons. First, agricultural decision-making involves material vulnerability: farmers rely on algorithmic recommendations that directly affect yield, income stability, and environmental outcomes, and erroneous advice may result in substantial financial loss. The vulnerability-centered definition, therefore, aligns closely with the decision context studied here. Second, Mayer's three-dimensional framework of ability, benevolence, and integrity provides a structured basis for decomposing and operationalizing perceived trustworthiness within a decision-support system.

We acknowledge that Mayer et al.'s model was originally developed to explain interpersonal and organizational trust. However, recent research has extended this framework to technological and AI contexts. For example, Bedué [44] empirically examines trust formation in AI systems using Mayer et al.'s framework and demonstrates that ability, benevolence, and integrity remain relevant in technological contexts, while also interacting with AI-specific considerations such as explainability, data quality, certification, and governance. In such socio-technical systems, these dimensions are attributed not to a single human actor but to the broader system—comprising, for example, the recommendation algorithm, the data and measurement pipeline it relies on, and the organization responsible for deployment and oversight. Consistent with prior work in automation and human-AI interaction, trust further depends on predictability, transparency, and users' ability to form coherent mental models of system behavior (Lee & See [43] and Hoff & Bashir [45]). Accordingly, we interpret and operationalize Mayer's dimensions within a socio-technical framework grounded in empirically observed farmer perceptions and practical agricultural decision-making conditions. Specifically, in our context, ability captures perceived agronomic competence of the recommendation system, benevolence captures perceived alignment with farmers' interests and practical constraints, and integrity reflects perceived transparency and consistency in how objectives and trade-offs are formulated and communicated.

In this study, we build upon the mechanisms outlined by Mayer et al. [17] and broadly hypothesize that several key factors play a critical role in shaping farmers' trust in AI-based agricultural management systems, with particular emphasis on their trust in AI agents. To explore this, we use the previously mentioned three-dimensional trust model as the foundation to develop a farmer-specific trust model, grounded in the survey data and subsequent analysis.

The first dimension of our model focuses on the AI agent's ability, emphasizing its competence in achieving desirable outcomes. To quantify this, we derived Eq. (7) to approximate farmers' trust in the AI agent's ability to provide effective fertilization recommendations. While the five recommendations presented in the survey resulted in comparable corn yields, all exceeding the U.S. average of 8,649 kg/ha (137.8 bu/ac) reported by the USDA in 1999 [48], yield (Y) remains the most direct measure of agronomic success, as it directly impacts farmers' net incomes. Therefore, we adopted 8,649 kg/ha as the baseline value in Eq. (7).

$$\text{ability} = \frac{Y}{8649} * \frac{1}{cosh(0.1 * (\sum_t N_t - 196))} * \frac{0.5}{(|N_F - 2.5|)} * \frac{O_F + 1}{N_F + 1} \quad (7)$$

Other key indicators, including fertilizer usage ($\sum_t N_t$), fertilization frequency (N_F), and the total number of fertilizer applications within the optimal window (O_F), also serve as measures of the AI agent's competence, as they reflect how well its recommendations align with standard farming practices and influence farmers' trust. According to our survey, appropriate fertilizer usage ranged from 150 to 200 lb/acre (approximately 168 to 224 kg/ha), with an average of 196 kg/ha. To capture the idea that farmers are less likely to trust recommendations that significantly deviate from this average, we applied the hyperbolic cosine function cosh in Eq. (7) to smoothly penalize such deviations. A scaling factor of 0.1 was introduced to ensure that small deviations would not result in disproportionately large reductions in trust. Additionally, we set a baseline fertilization frequency of 2.5 in Eq. (7), reflecting survey responses that indicated two to three applications per growth cycle as ideal. Lastly, the optimal fertilizer application window was defined as April to May, based on the majority of survey participants' responses.

The second dimension, benevolence, reflects whether farmers perceive AI agents as beneficial to their farming practices and aligned with their interests. In this study, we evaluated benevolence by examining the total amount of nitrate leaching ($\sum_t L_t$) resulting from the AI-generated recommendations, using it as a proxy for environmental impact. While most survey participants expressed limited concern about nitrate leaching — and some even viewed efforts to reduce it negatively — they generally believed that overemphasizing environmental benefits could negatively affect their income. As shown in Table 4, recommendations that prioritized environmental outcomes were typically less preferred. However, a subset of participants noted that effective nitrate control could enhance their trust in AI-driven fertilization management, as it contributes to long-term agricultural sustainability.

Table 7
Trust scores for AI-generated recommendations.

	Trust score
AI Recom 1	0.0002
AI Recom 2	0.01
AI Recom 3	0.04
AI Recom 4	0.04

We set the nitrate leaching baseline at 0.14 kg/ha, which corresponds to the outcome of AI Recommendation 4 - a recommendation that accounted for labor costs but not nitrate leaching. This recommendation was the most preferred, selected by 44.4% of participants. Nitrate leaching levels below this baseline may imply that the AI agent over-prioritizes environmental concerns at the expense of farmers' practical needs, whereas levels significantly above it could indicate inefficient nitrogen fertilizer use, potentially compromising both productivity and environmental health. Based on this rationale, we formulated Eq. (8) to evaluate the AI agents' benevolence in generating fertilization recommendations.

$$\text{benevolence} = e^{-\left(\frac{\sum_t L_t - 0.14}{0.1}\right)^2} \quad (8)$$

The final dimension, integrity, captures farmers' perceptions of the transparency and fairness of a recommendation. It reflects whether farmers believe that the proposed agricultural management strategy is unbiased, honest, and openly designed. In this study, integrity is assessed based on the credibility and transparency of the recommendation source rather than on performance outcomes. We assume that both the AI-generated recommendations and the expert baseline recommendation are developed and presented by independent researchers following standard agronomic practices, without deceptive intent or conflicting incentives, and are accompanied by clear explanations of their decision-making rationale. Under these assumptions, the integrity dimension is fixed to its maximum value (integrity = 1) for all recommendations, including the expert recommendation.

In summary, we use Eq. (9) to calculate the overall trust score, which allows us to quantify trust and optimize recommendations,

$$\text{Trust Score} = \text{ability} * \text{benevolence} * \text{integrity} \quad (9)$$

It is important to note that while the benevolence and integrity components are constrained to values between 0 and 1.0, the ability score may slightly exceed 1 due to the yield component in Eq. (7). By allowing this component within the ability dimension to surpass 1.0, we reflect this weighting and emphasize the yield's dominant role in shaping farmers' trust in AI-generated fertilization policies. This design decision aligns with findings from our survey (Section 4.2), which indicate that farmers prioritize corn yield (40.24%) more heavily than fertilizer amount (23.91%), fertilization frequency (19.02%), or environmental impact (16.83%).

Table 7 presents the trust scores for AI-generated recommendations based on the three-dimensional trust model, Eq. (9), developed in this study. Although all scores are relatively low, reflecting the AI agent's approach to intelligent agricultural management without considering farmers' trust, the order of these scores closely matches the trust rankings reported by survey participants (Table 6). AI Recommendations 3 and 4 receive the highest trust scores, primarily due to their optimal fertilization schedules and well-balanced nitrate leaching levels. AI Recommendation 2 earns a moderate trust score, indicating a reasonable but less optimal alignment with farmers' expectations and perceived interests. Moreover, AI Recommendation 1, however, receives the lowest trust score, highlighting a significant disconnect with farmers' priorities, as evidenced by its ranking in Table 6.

Interestingly, the expert recommendation receives a low trust score of 0.01, which contrasts with its relatively high ranking based on participants' survey responses (Table 6). This discrepancy can be attributed to the fact that participants' trust in the expert recommendation was likely influenced more by a general confidence in human expertise than by a detailed evaluation of the recommendation itself. In other words, while participants may have rated the expert recommendation highly due to an overarching trust in human knowledge and experience, they may not have closely examined the specific content of the recommendation when assessing its trustworthiness.

Such behavior is consistent with findings from previous studies. Despite demonstrated advantages of AI agents, individuals often exhibit a reluctance to rely on algorithmic decision-making — a phenomenon known as algorithm aversion. This refers to the tendency to lose trust in algorithms after observing even minor errors, even when those algorithms outperform human decision-makers on average [49]. This pattern highlights a key distinction: whereas trust in human experts often draws from generalized, affective, or identity-based beliefs, trust in AI-generated recommendations tends to require more deliberate, evidence-based justification.

Importantly, a trust-aware algorithm should not be conflated with a deceptive or strategically manipulated system. The trust score proposed in this study quantifies perceived trustworthiness from the farmers' perspective rather than providing a normative ethical evaluation of the AI system. A legitimate concern is that adjusting recommendations to increase acceptability could undermine integrity if such adjustments conceal relevant trade-offs or knowingly distort agronomic facts. In our framework, however, trust-awareness is implemented transparently through an explicit multi-objective optimization process. The agent does not distort agronomic outcomes or suppress environmental consequences; rather, it balances technical performance with empirically grounded, user-aligned criteria derived from farmer survey data. Integrity, within our model, depends on transparent disclosure of objectives, weighting schemes, and trade-offs. When these elements are openly communicated, incorporating trust into optimization enhances socio-technical alignment without compromising algorithmic integrity.

Beyond enhancing acceptability, this process increases the proximity between algorithmic behavior and farmers' own decision-making practices. Trust in automated systems depends not only on outcome performance but also on users' ability to form coherent mental models of system reasoning (Lee & See [43] and Hoff & Bashir [45]). Human-AI interaction guidelines further emphasize that systems should align with users' workflows and expectations to support calibrated trust (Amershi et al. [6]). Farmers typically rely on bounded heuristics, such as preferred fertilization windows, acceptable frequency ranges, and economically justified nitrogen levels, when managing uncertainty. When AI-generated recommendations deviate substantially from these experiential norms, they may appear misaligned or opaque, even if technically optimal. By incorporating empirically observed priorities and operational constraints into optimization, the resulting policies exhibit greater procedural similarity to established farming practices, thereby reducing cognitive distance between computational reasoning and situated agricultural judgment. In this sense, trust-aware optimization represents value alignment and decision-structure alignment rather than manipulation of outcomes.

5. Trust-aware intelligent agricultural management

In this section, we aim to address the challenge of agricultural nitrogen management by jointly considering two primary, and potentially conflicting, objectives: (1) maximizing agronomic performance, which includes optimizing crop yield, fertilizer usage, and nitrate leaching control, and (2) maintaining a high level of trust in the AI agent's decision-making. These objectives can be in tension; for instance, strategies that overly prioritize nitrate control may conflict with farmers' preferences and reduce trust, while trust-enhancing approaches may compromise task efficiency.

To examine the specific influence of the trust objective on policy learning, we set $w_4 = 0$ in the reward function (Eq. (6)), where w_4 represents the labor-cost penalty associated with fertilization frequency. Fertilization frequency (N_F) is already incorporated into the trust model (Eq. (7)) as a key determinant of perceived ability and alignment with farmer practices. Retaining the labor-cost term in the agronomic reward would therefore cause fertilization frequency to influence both the primary reward and the trust objective simultaneously, introducing potential confounding between economic efficiency and trust alignment. By removing the labor-cost penalty (i.e., $w_4 = 0$), fertilization frequency is no longer explicitly penalized in the reward function, thereby reducing overlap between objectives and allowing its influence within the MORL framework to be more clearly attributable to the trust component. Accordingly, the problem is reformulated as a multi-objective optimization problem, and Problem 1 is revised as follows.

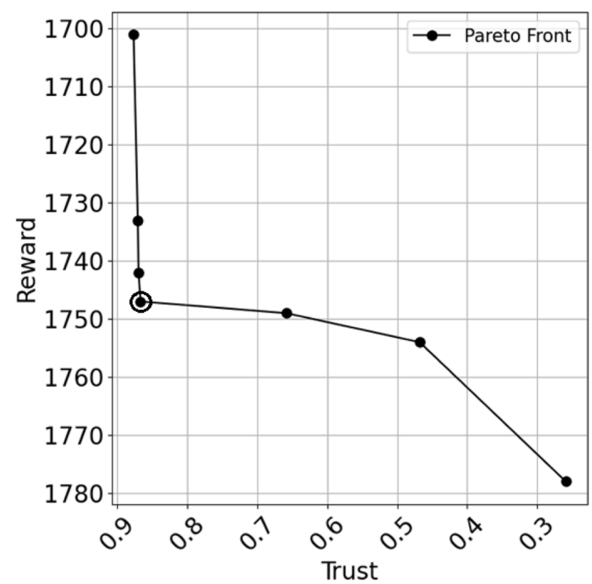

Fig. 4. Pareto fronts. The circled point represents the selected policy.

Problem 2. A POMDP, as defined in Section 2.1, models the agricultural nitrogen fertilizer management task, where the environment dynamics are simulated using the Gym-DSSAT framework. A primary reward function (Eq. (6)) evaluates agronomic performance by jointly considering crop yield, fertilizer usage, and environmental impact. In this formulation, the labor cost component is excluded, allowing the focus to remain on economic return and environmental sustainability. Additionally, a trust model (Eq. (9)) quantifies farmers' trust in the AI agent's recommendations. The objective is to learn an optimal policy that generates trust-aware fertilization strategies by simultaneously maximizing the cumulative reward and the trust score.

To address this problem, we adopted an MORL approach using DQN, as detailed in Section 2.3, to simultaneously maximize agricultural rewards and farmers' trust in fertilization recommendations. MORL allowed us to systematically explore and balance these competing objectives by generating a set of Pareto-optimal solutions. Upon convergence, the MORL framework produced a Pareto front, which is a collection of non-dominated policies representing optimal trade-offs between objectives. To select a final policy from the Pareto front, we employed an explicit preference-weighting strategy, assigning equal importance (50:50) to agricultural rewards and farmers' trust. This balanced approach ensured that the selected recommendations emphasized both agronomic effectiveness and perceived trustworthiness. Fig. 4 shows Pareto fronts illustrating the trade-offs between reward and trust, and the circled point represents the selected policy.

We first re-studied the intelligent agricultural management subject to the normal weather of 1999, as shown in Fig. 2. After the optimal policy was learned by the AI agent, a fertilization recommendation could

Table 8
Agricultural outcomes from different recommendations.

Recommendation	Trust-aware	Expert	Trust-agnostic
Total reward	1747	1697	1763
Yield (kg/ha)	9245	9248	9248
Nitrogen input (kg/ha)	190	224	180
Nitrate leaching (kg/ha)	0.12	0.26	0.09
Fertilization frequency	2	1	8
The number of nitrogen applications in April and May	2	1	1
Trust score	0.867	0.01	0.0002

be generated. We referred to this AI-generated recommendation as trust-aware because the agent considered farmers' trust during the learning process. As a comparison, we chose AI Recommendation 1 introduced in Section 3.2 since it has the same agronomic objectives: maximizing crop yield while minimizing fertilizer usage and nitrate leaching. However, the AI agent didn't consider farmers' trust when learning and generating AI Recommendation 1, so it was referred to as trust-agnostic. Additionally, we added the expert recommendation from Gym-DSSAT into the comparison. Table 8 summarizes the agricultural outcomes from three different fertilization recommendations.

It is important to note that the absolute values of nitrate leaching reported in Table 8 are relatively low. This outcome is attributed to the specific environmental and management conditions modeled in the simulation. First, the study simulates a rain-fed agricultural system typical of Iowa, excluding artificial irrigation, which often accelerates leaching. Second, the simulations utilized weather data from 1999, a year characterized by normal precipitation patterns and a lack of extreme rainfall events that typically trigger significant leaching. Finally, the AI-generated policies are explicitly optimized to synchronize fertilizer application with crop uptake, thereby minimizing the excess nitrogen available for leaching. Consequently, these values reflect high uptake efficiency under specific hydrological conditions, and our analysis focuses primarily on the relative differences in environmental impact across the different management strategies.

The results presented in the table reveal that the trust-agnostic AI-generated recommendation achieves the highest overall reward among the three recommendation strategies evaluated. This recommendation suggests its strong technical effectiveness in optimizing the targeted agronomic and environmental outcomes. However, despite its superior reward, this recommendation receives the lowest trust scores. The primary reason for this low level of trust is its failure to incorporate farmers' preferences, practices, and perceptions. Specifically, the recommendation favors small but frequent fertilizer applications as a strategy to minimize nitrate leaching. While environmentally beneficial, this approach significantly increases the operational burden on farmers by requiring more field visits and extended working hours. Additionally, the fertilizer application dates proposed by this strategy often deviate sharply from farmers' traditional schedules, creating further misalignment with their expectations and routines and thereby exacerbating their reluctance to adopt the recommendation.

In contrast, the expert recommendation takes a much more conservative and familiar approach. It suggests applying fertilizer only once at the beginning of the corn growing season, aligning more closely with traditional farming practices. As a result, this strategy achieves moderately higher trust scores in simulation studies, reflecting a greater level of comfort and acceptance among farmers. However, this approach also results in greater fertilizer use and consequently higher levels of nitrate leaching, which undermines its environmental sustainability and reduces its overall reward. Ultimately, despite being more trusted, the expert recommendation performs the worst in terms of reward among the three strategies.

The trust-aware AI-generated recommendation offers a compelling middle ground by balancing technical performance with anticipated social acceptability. This strategy achieves a reward comparable to the trust-agnostic recommendation, while also yielding a substantially higher trust score than the expert recommendation. Although we did not conduct a direct survey to assess farmer responses to this specific recommendation, the trust model used to guide its design was developed based on survey data capturing farmers' behaviors and preferences. By leveraging this information, the trust-aware recommendation strategically calibrates both the frequency and timing of fertilizer applications to better align with typical farming schedules. It proposes application periods that coincide with times when farmers are usually active in their fields, thereby reducing perceived disruptions. This contextual alignment is intended to ease implementation and foster a sense of familiarity and control—factors associated with increased trust and the likelihood of adoption.

6. Conclusion, limitations, and outlook

In this work, we introduced a mathematical trust model and integrated it into an MORL framework to generate optimal agricultural management policies on a crop simulator. Our primary aim was to maximize agricultural productivity while explicitly incorporating farmers' trust considerations. We conducted a detailed survey involving 71 farmers, primarily from Iowa, resulting in 54 high-quality, usable responses. Using these responses, we developed and validated a quantitative trust model based on the three-dimensional trust model of ability, benevolence, and integrity. This model was then incorporated directly into the reinforcement learning training process, enabling real-time trust feedback alongside traditional performance metrics such as crop yield, nitrogen use, and nitrate leaching. Simulation results demonstrated that the generated trust-aware optimal policies effectively balanced agricultural performance and farmer trust compared to expert-derived and trust-agnostic policies.

However, several limitations should be acknowledged. First, although we leveraged survey-derived trust models and farmer-reported practices to inform policy selection, we did not conduct follow-up surveys or behavioral validation studies to directly assess farmers' trust responses to the specific recommendations generated by the proposed framework. As a result, the trust outcomes reported in this study are model-based estimates rather than empirically observed post-intervention responses.

Second, the simulation experiments did not explicitly incorporate historically observed extreme weather events. Because such events can substantially influence crop performance, fertilizer efficiency, and risk perceptions, excluding them may limit the generalizability and robustness of the learned policies under climate extremes.

A final limitation concerns the interaction between the underlying algorithmic architecture and the trust assumptions adopted in this study. Although we utilized a POMDP formulation to ensure operational feasibility under partial observability, we assumed that the system's technical integrity remained constant and maximal ($Integrity = 1$). Under this assumption, the partial observability of the decision process did not explicitly influence the quantified trust score in our current model, as farmer trust was evaluated solely based on agronomic outcomes (Ability) and alignment with practical preferences (Benevolence). In real-world deployments, however, the probabilistic nature of POMDP belief-state estimation and the associated uncertainty may influence users' perceptions of system transparency and reliability. The internal state estimation process of a POMDP can be difficult for non-expert users to interpret, which may affect how the system's recommendations are perceived and trusted.

To address these limitations, future work should focus on expanding the survey to encompass a larger and more diverse group of participants, targeting over 200 respondents. We also plan to explicitly incorporate climate variability by leveraging historically observed extreme weather events, such as the 1983 heatwave and the 1988 drought in Iowa [4]. Integrating these coupled and realistic climate scenarios will enable a more rigorous evaluation of the trust model under environmentally stressed conditions and improve the robustness of the learned policies. In addition, we intend to conduct direct empirical evaluations of farmers' trust in specific AI-generated recommendations. Such follow-up assessments will provide stronger validation of perceived trustworthiness and practical acceptability, while enabling iterative refinement of the trust model based on feedback tied to individual recommendations.

Finally, future research should therefore relax the assumption of perfect integrity and explicitly examine how algorithmic architecture, uncertainty representation, and transparency mechanisms influence farmer trust. In particular, developing explainable artificial intelligence (XAI) interfaces that translate belief states and environmental uncertainties into interpretable information for users will be an important step toward bridging the gap between advanced probabilistic decision models and effective human-AI collaboration in agricultural management.

Consequently, beyond agricultural management, the proposed trust-centric MORL framework holds broader potential for domains in which user trust and acceptance are critical, such as personalized healthcare and autonomous transportation. By explicitly prioritizing trust in AI development, this line of work advances the design of decision-support systems that are more closely aligned with human values, expectations, and practical needs.

7. Ethical approval and informed consent statement

All experimental procedures involving human participants were conducted in accordance with the relevant institutional and national guidelines and regulations. Ethical approval for the study was obtained from the Institutional Review Board (IRB) of the University of Iowa under approval reference number 202405307, dated 05/22/2024.

All participants provided written informed consent prior to their inclusion in the study. The privacy rights of all participants have been fully observed, and all data were anonymized to maintain confidentiality.

8. Glossary

DQN – Deep Q-Network A type of reinforcement learning algorithm that combines Q-learning with deep neural networks. DQNs use neural networks to approximate the Q-value function, enabling agents to learn optimal policies in high-dimensional or continuous state spaces.

HAII – Human-AI Interaction A field of study focused on the design, understanding, and evaluation of systems where humans and artificial intelligence agents interact. It encompasses aspects such as usability, trust, transparency, and collaborative decision-making between humans and AI systems.

MORL – Multi-Objective Reinforcement Learning A reinforcement learning approach that simultaneously optimizes multiple, but often conflicting, objectives, such as maximizing crop yield while maintaining farmer trust.

POMDP – Partially Observable Markov Decision Process A framework for decision-making under environmental uncertainty, where an agent must act based on limited observations of the true system state.

RL – Reinforcement Learning A machine learning paradigm in which an agent learns to make decisions by interacting with an environment, receiving rewards or penalties based on its actions. The goal is to learn a policy that maximizes cumulative reward over time.

CRediT authorship contribution statement

Zhaoan Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Data curation, Conceptualization; **Wonseok Jang:** Writing – review & editing, Methodology, Formal analysis, Data curation, Conceptualization; **Bowen Ruan:** Writing – review & editing, Funding acquisition, Conceptualization; **Jun Wang:** Writing – review & editing, Project administration, Funding acquisition, Conceptualization; **Shaoping Xiao:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Funding acquisition, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This material is based upon work supported by the National Science Foundation under grant numbers 2226936 and 2420405 and the U.S. Department of Education under grant number ED#P116S210005. Any opinions, findings, and conclusions, or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the National Science Foundation and the U.S. Department of Education.

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